5. Revised SOTA Positioning

M3 Research Project · Detection-Constrained Power Allocation for Cognitive ISAC Relays under Imperfect CSI (NCC 2027 submission)

Source: submitted NCC 2027 paper, Sec. I (related work and contributions), Sec. V and Sec. VI. Reference numbers follow the paper.

Related work

Work	What it does	What is missing for this problem
Hu et al. [3] (base work)	Cognitive-radio ISAC relay. Joint secondary power allocation and beamforming to maximise the secondary SAR under sensing-SINR constraints. Covers perfect CSI and bounded channel errors.	The received-signal model has additive Gaussian noise and no separate clutter component. Protection uses an SINR floor, not a detection probability.
Behdad et al. [4]; Rivetti et al. [5]	Clutter-aware target detection in cell-free massive MIMO ISAC. [4] also allocates power with a sensing-SINR metric. Both represent the SINR-floor approach.	SINR-type metrics are tied to detection only under the assumed statistical model, which is Gaussian disturbance.
Yoo et al. [7]	Robust beamforming for coherent distributed ISAC under RCS and phase-synchronisation uncertainty. Represents the worst-case-over-a-bounded-set approach.	A worst-case guarantee pays a rate cost as the error set grows (Fig. 2).
Conte et al. [8]; Zhang et al. [12]; Liu et al. [13]	Compound-Gaussian clutter: NP/CFAR detection [8], ISAC detection-performance analysis and input-distribution design [12], clutter-aware ISAC survey [13].	Compound-Gaussian clutter is not used to constrain joint ST power, filter and relay allocation in a cognitive relay.
An et al. [9]; Guo et al. [10]; Mazahir & Tabassum [11]	Detection-probability vs rate trade-offs, including imperfect CSI [10], and detection-aware rate-CRB allocation [11].	Different architectures and allocation variables.

Position of this work

The change relative to [3] lies in two places:

- **the disturbance model:** compound-Gaussian clutter with statistical CSI errors;
- **the protection criterion:** each target's calibrated detection probability, $P_D \geq \beta$ at a fixed P_{FA}, replaces a prescribed sensing-SINR floor.

The requirement is averaged over the clutter and channel-error distributions. That makes this design a different point on the rate-protection trade-off from worst-case designs over bounded uncertainty sets.

How the results support this positioning

1. **The detection criterion is the source of the gain.** The component study keeps the allocation procedure fixed and changes only the radar-protection constraint. The proposed constraint gives 8.43 bit/s/Hz, +39.6 % over the worst-case SINR rule (6.04).

2. **Modelling the clutter matters.** A detection constraint built on a Gaussian disturbance model protects only about 68 % of channels, and a nominal SINR floor protects about 38 %.
3. **SINR floors over-protect.** At full observed protection (Fig. 3), the proposed design gives about 8.4 bit/s/Hz, against about 7.3 for the best benchmark (+15 %). The benchmarks reach full protection only by raising an SINR floor. Fig. 4 shows the result: their P_D sits near 1, while the proposed design sits close to the required 0.9.
4. **Advantage under CSI errors.** The gap widens as ε grows: +76 % over Hu robust at $\varepsilon = 0.01$, and 6.57 against 4.43 bit/s/Hz at $\varepsilon = 0.03$. At $\varepsilon \leq 1.5 \cdot 10^{-3}$, a tuned SINR floor is within 3 %, so the advantage is specific to non-negligible CSI errors.
5. **Robust to the clutter shape.** The protected rate changes by about 0.6 % between $\nu = 50$ and $\nu = 0.3$, with all channels protected.

Boundaries of the claim (stated in the paper)

- **Error-averaged requirement.** The requirement is not met on every error draw: 71 % of draws, against 100 % for the worst-case designs. On the worst-case SAR criterion, Hu robust is higher (2.86 against 2.30 bit/s/Hz).
- **Scope of the conclusions.** They apply to the adopted Gaussian-leakage detection approximation and to the evaluated channels.
- **Future work.** Error-controlled probability evaluation, and a probabilistic constraint on channel-error violations.

One-sentence positioning

For cognitive ISAC relays under compound-Gaussian clutter and imperfect CSI, the method replaces the sensing-SINR floor of [3] with a calibrated per-target detection constraint, and the results show that this admits more secondary traffic at the required error-averaged detection level than SINR-floor and worst-case designs.